4 DIGITAL COMMUNICATION

Our world is connected by wired and wireless digital communication systems. Data flows around these systems, carrying information about our personal and work lives and providing us with entertainment and news. Digital devices exchange data and communicate with each other and with larger systems.

Understanding the way in which devices communicate will help you to understand why things do not work as expected and will enable you to stay connected to the streams of data that drive our world.

By the end of this chapter, you should be able to select the most appropriate digital communication for a particular situation, after considering how each option would affect the quality of a connection.



LEARNING OBJECTIVES

- Know the range of ways that digital devices communicate: satellite, broadcast (TV, radio), wired (cable), wireless
- Know that digital devices can communicate device to device and by using networks: local area network (LAN), wide area network (WAN), personal area network (PAN), tethering
- Know the types of wireless communication: Wi-Fi, Bluetooth, GPS, 3G, 4G, infra-red (IR), near-field communication (NFC)
- Know the differences between Wi-Fi and Bluetooth and when each is best used
- Understand factors influencing the speed and volume of data transfer
- Understand the benefits and drawbacks of wired versus wireless communication in local networks
- Understand the significance of bandwidth and latency, and their impact on the 'user experience'
- Understand the features of broadband, mobile broadband and cellular networks

SPEED AND VOLUME OF DATA TRANSFER

As we transfer more and more data, it becomes more important that we understand how to increase the **speed** at which we can transfer it. If we do not, transferring more data will mean that transfer times will increase.

BANDWIDTH

SUBJECT VOCABULARY

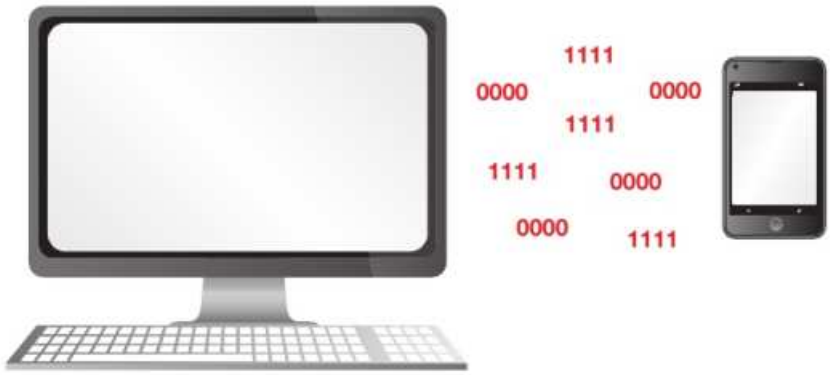
bandwidth the number of bits that can be carried by a connection in one second

The speed at which devices can transfer data depends on the **bandwidth** of the connection.

You can think about a data connection as being like a water pipe. In the same way that a large water pipe can carry a large amount of water, a connection with a large bandwidth can carry a large amount of data.

IMPACT ON USER EXPERIENCE

A higher bandwidth means that more data can be transferred every second. This makes uploads and downloads faster. It also makes it possible to do things that require lots of data to be transferred in short amounts of time, such as multiplayer online gaming or high-definition video streaming.



▲ **Figure 4.1** Binary data is stored within and transferred between digital devices

SUBJECT VOCABULARY

buffer an area of memory used to temporarily store data, especially when streaming video

When streaming video, all of the data does not need to be downloaded before playback can start. Instead, a portion of the video data is stored temporarily in an area of memory called a **buffer**. The video will not start until there is enough data in the buffer to play a few seconds of video. While those few seconds are playing, more data is downloaded to fill up the buffer. If the buffer is empty, there is no more video to play and it will pause until more data is downloaded. To avoid the buffer becoming empty, data must be constantly downloaded into the buffer, filling it up at a rate faster than it is emptied.

Imagine a glass of water being repeatedly filled up from a big bottle. The bottle is like the full video data file and the glass is like the buffer that holds the data ready to be used.



▲ **Figure 4.2** A visual representation of streaming

LATENCY

SUBJECT VOCABULARY

latency the amount of time it takes to send data between devices
domain the name used to identify a web server

As well as bandwidth, the speed of data transfer also depends on **latency**. Latency is the delay in the time it takes to send data between devices. If you think again about a data connection acting like a water pipe, latency is the average time that it takes for a drop of water to flow through the pipe.

You can identify latency by 'pinging' a **domain**. When you ping a domain, you send a packet of data to a server and the packet of data is immediately returned by the server to the originating device. The 'ping time' is the amount of time it takes for the packet of data to make the return trip.

SKILLS INTELLECTUAL INTEREST AND CURIOSITY
EXECUTIVE FUNCTION
REASONING
PROBLEM SOLVING

ACTIVITY

▼ PINGING DOMAINS

Using a command line interface (Terminal in MacOS and Linux or Command Prompt in Windows), type 'ping [domain]' to ping the following domains:

- gov.au (by typing 'ping gov.au')
- newsfirst.lk
- gov.uk

Note the ping time for each domain. Why are they different?

SKILLS INTELLECTUAL INTEREST AND CURIOSITY

ACTIVITY

▼ SPEED TESTING

Carry out a speed test on the internet connections of a fixed line broadband device, such as a laptop or PC, and a mobile broadband device, such as a smartphone.

IMPACT ON USER EXPERIENCE

In online gaming, the game will play smoothly if the bandwidth is adequate. However, if the latency is high, events in the game will **lag** and the game will not seem responsive to the player's commands. When watching live television, high latency will result in a delay between the real-time events and the video being received for playback.

GENERAL VOCABULARY

lag move slowly or fail to keep up

FACTORS THAT AFFECT SPEED AND VOLUME OF DATA TRANSFER

When devices transfer data, they can be affected by many factors that stop bits from reaching their destination. These bits then have to be sent again, which slows the overall data transfer rate.

TRANSFER METHOD

Wireless methods have to work on a limited number of **frequencies**. In comparison, copper cable can carry more frequencies than wireless methods. This means that cabled methods can have more bandwidth available to them than wireless methods.

SUBJECT VOCABULARY

frequency the waveband at which a radio signal is transmitted

HINT

Some older wired methods have less bandwidth than newer wireless methods.

INTERFERENCE

Other electromagnetic signals disrupt or interfere with wired and wireless signals. For example, interference can be caused by signals from wireless devices, wireless routers and appliances emitting electromagnetic fields like fridges and microwave ovens. Cabled connections can be **shielded** from this interference by having the wires wrapped in a thin layer of metal.

GENERAL VOCABULARY

shielded protected

BLOCKAGES

Walls and furniture reduce the strength of wireless signals. This reduces the available bandwidth.

DISTANCE

The strength of a wired or wireless signal is reduced as the distance that it has to travel increases.

DEVICE-TO-DEVICE COMMUNICATION

Devices can connect directly to each other using wired or wireless methods. This is called device-to-device communication. Table 4.1 shows some examples of device-to-device communication.

SUBJECT VOCABULARY

minijack a plug and socket widely used for analogue audio signals in portable devices
HDMI (High-Definition Multimedia Interface) used to transmit video and audio data



▲ Figure 4.3 Device-to-device communication

▼ Table 4.1 Examples of device-to-device communication

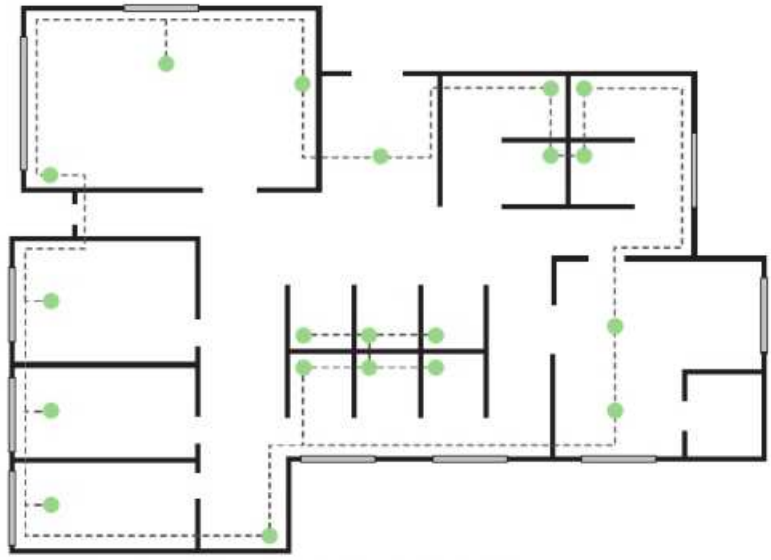
DEVICE 1	DEVICE 2	CONNECTION BETWEEN DEVICES 1 AND 2	USE
Temperature sensor	Air conditioner	Wired	To turn on the air conditioning when the temperature is too high
Smartphone	Headphones	**Minijack**	To play music from the smartphone on the headphones
Laptop	External hard drive	USB	To transfer files
Camcorder	Monitor	**HDMI**	To operate as a security camera
Games controller	Games console	Bluetooth	To control a game

NETWORK COMMUNICATION

When two or more computers are connected, a network is created. There are four major types of network.

LOCAL AREA NETWORK (LAN)

A LAN is a network that connects digital devices that are in a small geographical area, like a building or group of buildings that are close to each other.



▲ Figure 4.4 LANs are found in homes, schools and office buildings

WIDE AREA NETWORK (WAN)

A WAN is a network that is spread over a large geographical area. WANs are often used to connect different buildings owned by national and international businesses, law enforcement agencies, health and education organisations and government departments. Some organisations launch their own satellites to provide connectivity for their own global WANs.

WANs often use connectivity provided by a third-party telecommunications company, often linking LANs together through the internet. Because of their wider reach, WANs often have slower transfer speeds than LANs.

DID YOU KNOW?
The internet is the largest example of a WAN.



▲ **Figure 4.5** The internet is the largest example of a WAN

PERSONAL AREA NETWORK (PAN)

A PAN is a group of connected devices that are all near an individual user. For example, a user could connect their smartwatch to their smartphone, which is connected to their laptop and home cinema speaker system. Devices in a PAN can either be connected to each other directly or connected through access points (see page 84 for more information about access points).

When a PAN only uses wireless connectivity, it can also be referred to as a WPAN (**W**ireless **P**ersonal **A**rea **N**etwork). However, the general term PAN is more commonly used to refer to all types of PAN.

TETHERING

SUBJECT VOCABULARY

tethering connecting a host device that uses a mobile broadband connection with other devices so that they can use the host's broadband connection

Tethering is the process of connecting a host device, such as a smartphone or a tablet device, that uses a mobile broadband connection with one or more other devices. This enables the other device or devices to share the host device's broadband connection.



▲ **Figure 4.6** Devices can be tethered using wired or wireless connectivity

SUBJECT VOCABULARY

service agreement contract

Mobile phone network providers can enable or disable tethering as part of the **service agreement**. Some network providers charge more for this feature to be enabled.

THE WAYS IN WHICH DIGITAL DEVICES COMMUNICATE

The methods that digital devices use to share data and some common uses of these methods are shown in Table 4.2.

▼ **Table 4.2** Ways in which digital devices communicate and their common uses

Method	Technology	Use
Satellite	Radio waves	GPS, television, telephone, military
Broadcast	Radio waves	Television shows, radio shows
Wired	Electrical signal	Networking, connecting peripherals
Wireless	Radio waves	Networking, connecting peripherals

SATELLITE COMMUNICATION



▲ **Figure 4.7** A communication satellite

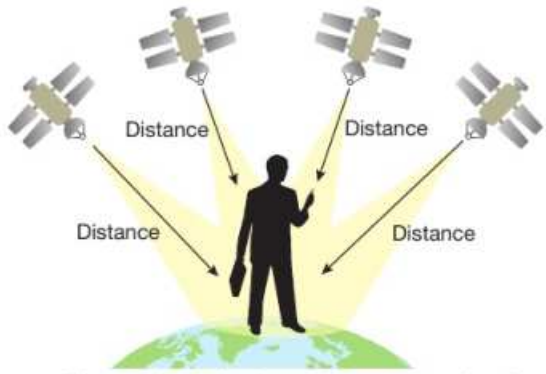
Satellites transmit data to and receive data from digital devices. Digital devices use antennae to receive the radio signals that satellites transmit.

The benefit of satellite communication is that the number of satellites means that the system is always available. It also cannot be affected by power shortages.The drawback is that satellite signals do not pass through solid objects. This means that they will not work in areas with tall buildings or in tunnels. Signals can also be affected by atmospheric weather conditions such as heavy snow or rain.

GPS

Satellite communications are used for GPS. Navigation aids make use of GPS signals to calculate the exact location of a device. GPS signals are sent from a network of 24 satellites orbiting the Earth. At any one time, a device will be within view of approximately 12 of these satellites. However, a view of only four satellites is required to calculate an accurate location, as illustrated in Figure 4.8.

For more information about navigation aids, see *Unit 1 Digital devices* (page 14)

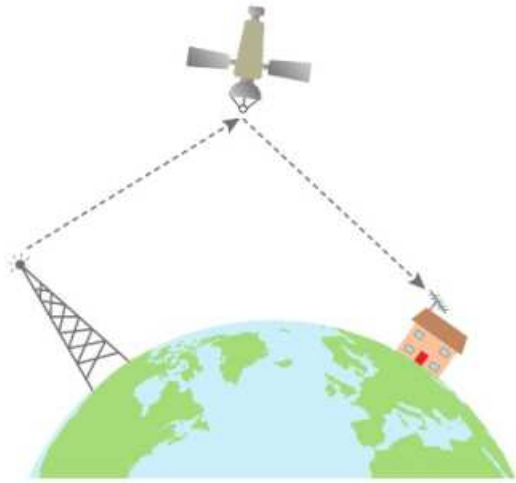


▲ **Figure 4.8** Only four satellites need to be in view to get an accurate location of a device using a GPS signal

TELEVISION

Digital Video Broadcasting (DVB) is the internationally accepted standard method of broadcasting digital television. DVB-S (Digital Video Broadcasting – Satellite) is one example of DVB. A video signal from the broadcaster is transmitted using a large antenna on Earth to one or more satellites, which

then broadcast the signal back down to Earth. A satellite television viewer will have an antenna installed, and this receives the signal and sends it to a set-top box. The set-top box decodes the signal and converts it so that it is ready to be sent to a television. Some televisions have decoders installed, so the antenna can be connected directly to the television rather than requiring a set-top box to decode the signal first.



▲ **Figure 4.9** DVB-S (Digital Video Broadcasting – Satellite)



▲ **Figure 4.10** A receiving antenna with a set-top box

DVB-S2 and DVB-S2X are newer digital broadcasting standards. They provide more functionality, such as High Definition Television (HDTV), interactive services and internet access.

TELEPHONE

Satellite communication is also used to allow people in remote areas to place voice calls using satellite telephones.





▲ **Figure 4.11** Satellite telephones use antennae to transmit data to (and receive data from) one or more satellites



▲ **Figure 4.12** Satellite phones are used in remote areas

MILITARY

The military in many countries use satellites for communication systems, such as the Global Command and Control System.

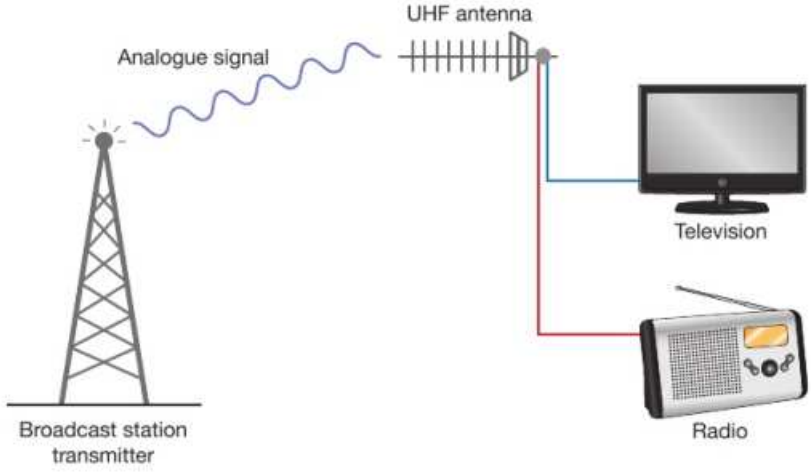


▲ **Figure 4.13** The Global Command and Control System provides military communication

BROADCAST COMMUNICATION

ANALOGUE TELEVISION AND RADIO

Transmitters broadcast television and radio signals that are received by a viewer's antenna. This antenna sends a signal through a wire to the television or radio receiver, which converts it into images and audio.



▲ **Figure 4.14** An analogue signal is received by an antenna and fed to different devices

DID YOU KNOW?

In some countries, analogue signals have been switched off and only digital signals are now broadcast. In January 2017, Norway became the first country to begin switching off the transmissions from its national FM radio stations.

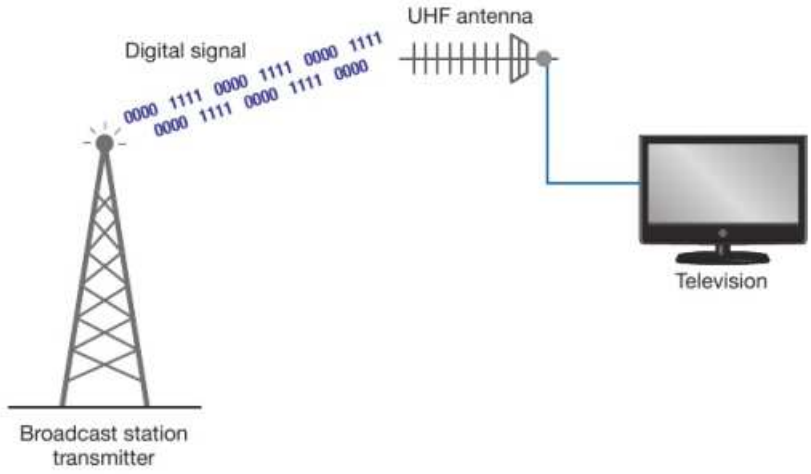
DIGITAL TELEVISION

DVB-T (Digital Video Broadcasting – **Terrestrial**) is a method of DVB where the transmitters are based on Earth, rather than in orbit as they are in DVB-S. To receive digital television broadcasts transmitted by DVB-T, viewers can use the same antenna that they use to receive analogue broadcasts. They do not need a special antenna.

DVB-T2 is a newer standard that provides more functionality, such as High Definition Television (HDTV) and interactive services.

GENERAL VOCABULARY

terrestrial on the Earth



▲ **Figure 4.15** Some televisions have decoders installed, so the antenna can be connected directly to the television rather than requiring a set-top box to decode the signal first

DIGITAL RADIO

DAB (Digital Audio Broadcasting) is used in Europe and the Asia Pacific region. It is broadcast in the same way as DVB. DAB provides more radio stations and can also carry text data that DAB receivers can display. The text data can include the time, name of the station and details of the music being played.

WIRED COMMUNICATION

SUBJECT VOCABULARY

Ethernet a network connectivity standard that provides a way for computers to communicate

Devices can use cables to communicate with each other via a wired connection. There are many different types of wired connection. Some are used for many different purposes, such as USB. Others are only used to meet one particular need, such as **Ethernet**.

Some of the uses for different wired connection types are given in Table 4.3.

▼ **Table 4.3** Common wired connection types and their uses

CONNECTION TYPE	USE
HDMI	Digital video connections
S/PDIF	Digital audio connections
Minijack	Personal headphones
USB	Storage transfer
Ethernet	Networking



▲ **Figure 4.16** An Ethernet connection

USB is a very common connection type. USB has been through a number of revisions, and each revision allows faster data transfer speeds. This development of standards is common with all types of connectivity. This progress is made necessary as digital devices become more complex in their features and functionality.

Ethernet allows a user to connect to wired networks. As Ethernet technology develops, the speed at which data can be transferred between devices is improved. Ethernet cables can be 100 metres long before the signals they carry start to lose quality.

WIRELESS COMMUNICATION

Devices can also use wireless connectivity to communicate with each other. Just as with wired communication, there are many different types of wireless connection.

HINT

The terms 'Wi-Fi' and 'internet access' are often used incorrectly to mean the same thing. Wi-Fi allows you to wirelessly connect to a network. That network may or may not be connected to the internet. For more information about networks, see pages 79–93.

DID YOU KNOW?

A laser transmitter in New Mexico, USA is used to beam a wireless signal 384,400 km to a satellite orbiting the moon. It can carry 19.44 megabits of data per second, which is more than enough to stream live television.

WI-FI

Wi-Fi is used in home and office networks. Some companies provide Wi-Fi access in towns and cities. Wi-Fi is a wireless technology used to connect devices to a network. That network can itself then be connected to the internet, so that devices connected to the Wi-Fi network can connect to the internet.

Wi-Fi uses the IEEE 802.11 specification of standards for wireless communication. The specification is revised regularly to take account of improvements in technology. Each revision to the specification is given a different letter or pair of letters at the end. The first version was IEEE 802.11**a**. Each revision of the technology improves the speed at which data can be transferred and increases the distance over which devices can connect.

SKILLS SELF-DIRECTION

ACTIVITY

▼ DIFFERENT REVISIONS OF WI-FI

Do some research to compare the different revisions of Wi-Fi. How do their speeds and range differ?

BLUETOOTH

SUBJECT VOCABULARY

pair connect two devices (usually only with each other)

Bluetooth is a type of wireless connectivity that lets devices connect over short distances. It cannot carry as much data as Wi-Fi. Bluetooth devices need to be **paired** with each other before they can communicate.

Bluetooth can be used to transfer small files between devices. It is used to connect devices such as smartphones and laptops to peripherals such as portable speakers, headphones, earphones, keyboards and mice.

▼ Table 4.4 Comparing Wi-Fi and Bluetooth

	WI-FI	BLUETOOTH
Range	Long ✓	Short ✗
Bandwidth	High ✓	Low ✗
Power	High ✗	Low ✓
Security	High ✓	Low ✗
Can connect multiple devices simultaneously	Yes ✓	Limited ✗ (usually have to be paired)

DID YOU KNOW?

Wi-Fi Direct is an alternative to Bluetooth. It is a low-power version of Wi-Fi that can connect two devices directly over a short range.

3G AND 4G

3G and 4G are sometimes referred to as mobile broadband. They are used to provide internet access to mobile devices such as smartphones and tablet devices when a Wi-Fi signal is not available. The G stands for 'generation', meaning that 4G is the fourth generation of mobile broadband technology. Future generations of the technology are planned to improve the speed and availability of the signal.

SKILLS SELF-DIRECTION
INTELLECTUAL INTEREST AND CURIOSITY

ACTIVITY

▼ 3G AND 4G

1 Compare the speeds, costs and availability of 3G and 4G in your location.

2 Research the next generation of this technology.

HINT

3G or 4G is **not** the same as Wi-Fi. 3G and 4G connectivity is provided by mobile phone companies. Some people say they 'have Wi-Fi' when they mean that they have a mobile broadband connection.

INFRA-RED (IR)

Infra-red signals cannot carry much data and only have a short range. Transmitters must have a clear line of sight to receivers, because this allows the signal to travel in a straight line between them without being blocked by solid objects like walls. The signal is also affected by sunlight. It is often used in remote-control devices such as television remote controls.

SUBJECT VOCABULARY

infra-red a type of electromagnetic radiation with a longer frequency than that of visible light

NEAR-FIELD COMMUNICATION (NFC)

NFC uses close proximity RFID (Radio Frequency Identification) chips. NFC is used in smartphones, payment cards and travel cards.

For more information about NFC and RFID chips, see *Unit 1 Digital devices* (page 30).

SKILLS SELF-DIRECTION

ACTIVITY

▼ COMPARING WIRELESS CONNECTIONS

Complete Table 4.5 by identifying the technology used by each type of wireless connection and some common uses for that type of connection.

▼ Table 4.5 Wireless connections: technology and uses

METHOD	TECHNOLOGY	USE
Wi-Fi	Radio waves	Home and office networks
Bluetooth		
3G and 4G		
Infra-red		
NFC		



▲ Figure 4.17 Infra-red is commonly used by remote controls to transmit signals

BENEFITS AND DRAWBACKS OF WIRED vs WIRELESS

The World Health Organization (WHO) says that current research findings suggest that exposure to wireless signals does not cause health issues. However, it also says that further long-term research about the effects on children using mobile phones for more than 10 years is necessary. This is because symptoms may take a long time to appear and young people are most vulnerable to radiation.

DID YOU KNOW?

The wavelength of Wi-Fi (12 cm) is at least 300,000 times longer than the wavelength of light (400–700 nanometres). The WHO generally considers anything above the wavelength of light to be harmless to humans.

▼ **Table 4.6** Comparing wired and wireless connectivity

	WIRED	WIRELESS
COST	Cables are cheap if purchasing for a small number of devices	• No need to buy cables • May need a wireless access point for multiple connections
SAFETY	Risk of tripping over cables	None (though some people are worried about the effects of radiation)
SPEED	Faster than wireless	Slower than wired
STABILITY	Less affected by interference than wireless	Affected by interference and obstacles
PORTABILITY	• Not portable as limited by connecting cables • May need signal booster if connection is more than 100 metres long	Portable within signal range
MESS	Can look untidy if lots of cables are used	Tidy
SECURITY	Most secure	Less secure than wired connection because it is easier to intercept a wireless signal
MAINTENANCE	Using cables and ports continuously over a long period of time may damage them	None

KEY POINT

The International Agency for Research on Cancer is part of the WHO. It classifies Wi-Fi as a possible human **carcinogen**. However, many things are possibly carcinogenic. For example, Wi-Fi shares this classification with coffee, carpentry, foam cups and pickled vegetables.

GENERAL VOCABULARY

carcinogen a substance that can cause cancer

HINT

Cellular network is another name for a mobile phone network.

SUBJECT VOCABULARY

Internet Service Provider (ISP) a company that provides customers with access to the internet
fibre optic cable a cable that sends data using light signals
copper cable a cable that sends data using electrical signals, which are conducted through copper wires

BROADBAND, MOBILE BROADBAND AND CELLULAR NETWORKS

Broadband networks provide fast access to the internet through a connection to an **Internet Service Provider** (ISP). They use the **fibre optic cable** or **copper cable** network. You will learn more about fibre optic cables on page 84.

Mobile broadband provides high-speed wireless connectivity using 3G or 4G technology to connect to the mobile phone network, which acts as the user's ISP.

CHAPTER QUESTIONS

SKILLS REASONING

1 a Explain **one** reason why some games console controllers use Bluetooth rather than infra-red to connect to the console. **(1)**

b State **one** way in which using infra-red in games console controllers could affect the experience of the person playing the game. **(1)**

SKILLS INTERPRETATION

2 Describe how video streaming works. **(3)**

SKILLS REASONING

3 Explain why streaming is more convenient for the user than downloading. **(3)**

SKILLS PROBLEM SOLVING

4 State **three** types of wired connection. **(3)**

SKILLS INTERPRETATION

5 Which **one** of these best describes the internet? **(1)**

A LAN
B WAN
C PAN
D VLAN

SKILLS REASONING

6 Explain why global games companies use games servers in multiple countries to ensure that the experience of users is not negatively affected when playing online games. **(3)**